

NOGOONNUUR, Mongolia

WMO: 442100

Lat: 49.6153N

Lon: 90.2433E

Elev: 7156

StdP: 11.27

Time Zone: 7.00 (E07)

Period: 07-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.8	-23.3	-33.0	1.1	-24.8	-29.8	1.3	-21.9	19.6	12.8	16.6	12.9	1.3	280	0.602

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	80.7	55.5	78.1	55.0	75.8	54.0	58.2	74.5	57.0	72.7	55.8	71.4	7.5	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
53.1	78.5	63.0	51.1	72.8	63.7	48.7	66.6	64.3	28.5	73.9	27.7	72.9	26.8	71.6	63.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
22.7	19.0	15.8	-28.6	85.7	6.9	3.0	-33.6	87.8	-37.6	89.6	-41.5	91.3	-46.5	93.5		
			-29.6	60.5	6.8	2.1	-34.4	62.0	-38.4	63.2	-42.2	64.4	-47.1	65.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	33.7	-7.5	2.3	24.6	41.2	50.5	63.0	65.7	61.8	49.8	35.5	15.8	0.2	(6)
(7)		DBStd	26.28	8.99	10.38	10.70	8.85	8.11	5.93	4.69	5.40	7.92	8.61	9.29	8.07	(7)
(8)		HDD50	7402	1781	1336	787	278	96	2	0	1	98	452	1027	1544	(8)
(9)		HDD65	11555	2246	1756	1252	714	453	102	48	126	458	914	1477	2009	(9)
(10)		CDD50	1459	0	0	0	14	110	391	485	366	90	3	0	0	(10)
(11)		CDD65	138	0	0	0	0	2	41	68	26	1	0	0	0	(11)
(12)		CDH74	926	0	0	0	0	23	310	408	167	18	0	0	0	(12)
(13)		CDH80	110	0	0	0	0	1	36	54	17	2	0	0	0	(13)
(14)	Wind	WSAvg	4.7	2.1	2.8	4.9	6.2	7.8	5.6	5.7	5.0	4.8	4.7	3.8	2.6	(14)
(15)	Precipitation	PrecAvg	6.5	0.1	0.1	0.2	0.4	0.6	1.1	1.7	1.2	0.5	0.3	0.3	0.2	(15)
(16)		PrecMax	9.4	0.4	0.2	0.6	0.7	1.1	3.1	4.3	2.6	0.9	0.6	0.5	0.4	(16)
(17)		PrecMin	4.4	0.0	0.0	0.0	0.2	0.2	0.3	0.5	0.3	0.2	0.0	0.1	0.0	(17)
(18)		PrecStd	1.3	0.1	0.0	0.2	0.2	0.3	0.6	0.9	0.6	0.2	0.2	0.1	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	23.2	33.8	53.3	67.0	77.3	83.7	85.7	82.1	76.3	61.0	49.9	30.1	(19)
(20)		MCWB	16.9	25.1	37.7	45.5	50.4	56.4	58.1	54.9	52.4	43.0	36.8	22.9	(20)	
(21)		2%	DB	17.4	28.0	49.3	63.4	71.9	80.4	81.0	78.0	71.4	56.1	40.1	22.8	(21)
(22)		MCWB	12.5	20.5	34.8	43.6	48.7	55.1	56.2	55.6	50.7	40.5	31.4	17.8	(22)	
(23)		5%	DB	12.1	23.0	46.0	59.7	68.8	77.5	78.4	75.4	67.3	53.0	35.2	16.9	(23)
(24)		MCWB	8.6	17.0	32.9	41.7	47.3	54.4	55.6	54.4	48.8	39.0	27.3	13.0	(24)	
(25)		10%	DB	6.6	18.4	42.5	56.2	65.3	74.8	76.0	73.0	63.7	49.7	30.6	11.9	(25)
(26)		MCWB	4.1	13.5	30.7	39.5	45.7	52.9	54.5	53.3	47.0	36.9	23.9	9.2	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.0	24.8	38.7	47.1	52.2	58.7	61.1	59.4	55.2	43.7	38.6	23.0	(27)
(28)		MDCB	21.9	33.1	51.5	64.4	72.7	77.9	77.3	73.8	71.1	59.0	47.6	29.8	(28)	
(29)		2%	WB	12.7	20.6	35.3	44.7	49.9	57.2	58.5	57.5	52.2	41.4	31.4	17.8	(29)
(30)		MDCB	17.3	27.8	47.7	61.3	69.6	75.0	75.3	72.3	67.9	54.8	39.3	22.8	(30)	
(31)		5%	WB	8.6	17.0	33.2	42.2	48.2	56.0	57.4	55.9	49.9	39.3	27.3	13.0	(31)
(32)		MDCB	12.2	22.9	45.3	58.3	67.1	72.7	72.7	71.3	65.4	52.4	34.6	16.6	(32)	
(33)		10%	WB	4.1	13.5	30.8	40.1	46.4	54.8	56.3	54.4	47.9	37.2	24.0	9.2	(33)
(34)		MDCB	6.6	18.3	41.9	55.0	63.7	71.0	71.4	70.3	62.8	49.1	30.4	11.9	(34)	
(35)	Mean Daily Temperature Range		MDBR	19.8	22.5	22.6	21.6	20.6	18.9	16.9	18.3	20.6	19.4	18.6	17.8	(35)
(36)		5% DB	MCDBR	21.5	24.0	23.6	25.2	24.1	21.9	20.3	21.2	24.1	22.6	22.1	20.6	(36)
(37)			MCWBR	18.1	18.5	14.4	13.5	11.4	8.4	8.3	8.8	11.4	13.4	15.7	16.7	(37)
(38)			MCDBR	21.7	24.3	22.5	23.2	21.9	19.3	17.2	19.2	22.6	22.2	21.3	20.3	(38)
(39)		5% WB	MCWBR	18.3	18.8	14.1	13.2	11.5	7.9	7.3	8.5	11.1	13.5	15.5	16.5	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.205	0.224	0.259	0.294	0.296	0.302	0.303	0.289	0.258	0.244	0.222	0.205	(40)	
(41)		taud	2.427	2.410	2.384	2.408	2.445	2.496	2.512	2.530	2.574	2.550	2.469	2.412	(41)	
(42)		Ebn at Noon	287	304	306	304	305	303	301	302	305	293	276	270	(42)	
(43)		Edh at Noon	18	24	30	34	34	33	31	30	25	21	17	16	(43)	
(44)	All-Sky Solar Radiation	RadAvg	493	828	1296	1734	2035	2077	2004	1786	1426	888	504	380	(44)	
(45)		RadStd	20	28	57	68	119	92	112	77	72	57	36	19	(45)	

Historical Trends

		DBAvg		Heating		Cooling			Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (6 neighbors)	N/A	N/A	-4.27	N/A	-1.37	-2.00	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page